WallCal

📊 Project Overview

Project Name: WallCal — Interactive Wall Calendar Component

Type: Reusable React/Next.js frontend component (no backend)

Summary: A polished, interactive calendar component styled after a physical wall calendar. Features a hero image visual anchor, date range selection with clear visual states, integrated multi-level notes (monthly, range, daily), localStorage persistence, and responsive design. Built as both a reusable component and a portfolio showpiece.

Timeline: 5–7 days for MVP, 10–14 days fully polished with creative extras

Solo Developer: Yes

Deliverables: GitHub repo, video demo, live Vercel deployment

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🎯 Product Vision

One-liner: A beautiful, wall-calendar-inspired React component that makes date range planning feel tactile and intentional.

Problem: Most web calendar components are functional but visually generic — they look like utility widgets, not experiences. The physical wall calendar has a charm (large imagery, spatial clarity, a notes margin) that digital calendars ignore.

Solution: WallCal translates the wall calendar aesthetic into an interactive web component with modern UX: smooth month transitions, intuitive range selection, integrated notes at three levels (month, range, day), and a hero image that anchors the visual identity of each month.

Why it matters: This component demonstrates that functional UI and beautiful design aren't mutually exclusive. It's reusable in any app that needs date planning (trip planners, project tools, event apps) and stands alone as a portfolio piece showcasing frontend craft.

👤 Target User

Primary: Frontend hiring managers and technical reviewers evaluating code quality, design sense, and interaction polish.

Secondary: Developers who might integrate this calendar component into their own applications for date-range-based planning (trip planning, event scheduling, project timelines).

User Characteristics:

- Expects smooth, responsive interactions on both desktop and mobile
- Notices visual details: hover states, transition timing, spacing consistency
- Values clean component architecture and readable code
- Uses modern browsers (Chrome, Firefox, Safari, Edge — latest 2 versions)

✨ Core Features

Feature 1: Wall Calendar Layout

- **Hero image panel:** Takes up approximately 55% of the calendar's total height on desktop, or full width stacked above the grid on mobile
- **Spiral binding visual:** A decorative element at the top edge of the calendar using CSS (repeating dots or semicircles) to evoke a real wall calendar — purely decorative, 24px height
- **Month/Year overlay:** Displayed on the bottom-right corner of the hero image in a semi-transparent badge — format: "2025 / JUNE" — white text, font-weight 700, with a `rgba(0,0,0,0.4)` backdrop blur
- **Desktop layout:** Single card, max-width 960px, centered. Hero image on top, calendar grid + notes below in a two-column layout (grid 60%, notes 40%)
- **Mobile layout (≤768px):** Full-width stacked — hero image → calendar grid → notes section, all vertically arranged
- **Card styling:** `border-radius: 12px`, `box-shadow: 0 8px 32px rgba(0,0,0,0.12)`, white background (dark mode: `#1a1a2e`)

Feature 2: Calendar Date Grid

- **7-column grid:** Monday through Sunday (matching the reference image which starts on Monday)
- **Column headers:** Three-letter abbreviations — MON, TUE, WED, THU, FRI, SAT, SUN
- **Current month days:** Rendered in the grid with correct weekday alignment
- **Overflow days:** Previous/next month days shown in the grid but at 30% opacity and non-interactive
- **Today indicator:** Current date has a subtle ring/outline in the accent color, 2px solid
- **Weekend styling:** SAT and SUN column headers and date numbers rendered in the accent color
- **Cell size:** Minimum 40×40px touch target on mobile, 48×48px on desktop
- **Hover state:** Background circle fades in at 10% accent color opacity, 150ms ease transition

Feature 3: Date Range Selector

- **Selection model:** Exactly one active range at a time per the entire calendar (not per month)
- **Selection flow:**
  1. First click on any date → sets it as the **start date**
  2. Second click on any date after the start → sets it as the **end date**, range is now complete
  3. Second click on a date before the start → that date becomes the new start, old start becomes end
  4. Clicking any date when a range is already complete → clears the range, clicked date becomes new start
  5. Clicking the existing start date when no end is set → clears selection entirely
- **Cross-month ranges:** If a range spans multiple months, both months show the relevant highlighted portion when navigated to
- **Visual states (5 distinct states):**
  - **Default:** No background, dark text

    - **Hover:** Light accent circle background (10% opacity)
    - **Start date:** Filled accent color circle, white text, left-rounded if range exists
    - **End date:** Filled accent color circle, white text, right-rounded
    - **In-range dates:** Accent color at 15% opacity as a continuous horizontal band connecting start to end, no gap between cells
    - **Today (unselected):** 2px accent border ring, no fill
- **Transition:** All state changes animate over 200ms ease-out
- **Accessibility:** Selected range announced via `aria-label` on the grid: "Selected range: June 5 to June 12"

Feature 4: Integrated Notes System

Three note types, one unified interface:

4a. Monthly Notes

- A text area always visible in the notes panel
- Placeholder text: "Notes for June..."
- Max length: 500 characters with a character counter shown at bottom-right of the text area
- Saved per month — switching months loads that month's note
- Auto-saves to localStorage on every keystroke (debounced 500ms)

4b. Range Notes

- Appears only when a date range is actively selected
- Displayed below the monthly notes area with a label: "Note for Jun 5 – Jun 12"
- Max length: 300 characters with counter
- Saved and keyed to the specific start–end date pair in localStorage
- If the same range is re-selected later, the note reloads

4c. Day Notes (Optional/Bonus)

- Triggered by double-clicking (desktop) or long-pressing (mobile, 500ms) on a single date

- Opens a small inline popover/tooltip with a text input, max 140 characters
- Dates with existing notes show a small dot indicator (4px circle) below the date number in the grid
- Saved per individual date

Notes panel styling:

- Background: `#f8f8f8` (dark mode: `#252540`)
- Border-left on desktop: `1px solid #e0e0e0`
- Text area: no visible border, subtle bottom-line underline style, font-size 14px, line-height 1.6
- Labels: font-size 12px, uppercase, letter-spacing 1px, 40% opacity text

Feature 5: Month Navigation

- **Navigation controls:** Left arrow (←) and right arrow (→) buttons flanking the month/year text above the calendar grid
- **Month/year display:** "June 2025" centered between arrows, font-weight 600, font-size 18px
- **Button style:** 36×36px circular buttons, icon-only (chevron), ghost style with hover fill at 10% opacity
- **Keyboard support:** Left/Right arrow keys navigate months when calendar component is focused
- **Animation:** Month transition uses a horizontal slide + fade: exiting month slides out 30px in the direction of navigation and fades to 0 opacity, entering month slides in 30px from the opposite side and fades to full opacity — 300ms ease-in-out
- **No infinite scroll:** Navigation is unrestricted (users can go to any month/year)
- **Hero image changes per month:** Each of the 12 months maps to a different hero image (use curated Unsplash landscape/nature images — one per month, static URLs)

Feature 6: localStorage Persistence

- **Storage key prefix:** `wallcal_`

- **Stored data:**
  - `wallcal_monthlyNotes` → `{ "2025-06": "text...", "2025-07": "..." }`
  - `wallcal_rangeNotes` → `{ "2025-06-05_2025-06-12": "text..." }`
  - `wallcal_dayNotes` → `{ "2025-06-05": "text..." }`
  - `wallcal_selectedRange` → `{ "start": "2025-06-05", "end": "2025-06-12" }`
  - `wallcal_currentMonth` → `"2025-06"`
  - `wallcal_theme` → `"light" | "dark"`
- **Write strategy:** Debounced writes, 500ms after last change
- **Read strategy:** Load on component mount, merge with defaults if keys missing
- **Error handling:** If localStorage is full or unavailable, component still functions (data just won't persist). Show a subtle toast: "Storage unavailable — notes won't be saved"

Feature 7: Dark Mode

- **Toggle:** A sun/moon icon button in the top-right corner of the calendar card
- **Behavior:** Toggles between light and dark themes. Preference saved to localStorage
- **Initial state:** Respects `prefers-color-scheme` media query on first load, then uses saved preference
- **Light palette:**
  - Background: `#FFFFFF`
  - Card surface: `#FFFFFF`
  - Text primary: `#1a1a1a`
  - Text secondary: `#6b6b6b`
  - Border: `#e5e5e5`
  - Notes panel bg: `#f8f8f8`
- **Dark palette:**
  - Background: `#0f0f1a`
  - Card surface: `#1a1a2e`

    - Text primary: `#e8e8e8`
    - Text secondary: `#9a9ab0`
    - Border: `#2a2a40`
    - Notes panel bg: `#252540`
- **Accent color:** Stays consistent across themes (derived from hero image or default `#E07A5F`)
- **Transition:** All color changes animate with `transition: background-color 300ms ease, color 300ms ease`

Feature 8: Hero Image Theme Tinting

- **Mechanism:** Each month's hero image has a pre-defined dominant color mapped in a static config object
- **Effect:** The accent color used throughout the calendar (range selection, weekend text, hover states, today ring) dynamically updates to match the hero image's dominant color when the month changes
- **Transition:** Accent color crossfades over 400ms when navigating months
- **Fallback accent:** `#E07A5F` (warm terracotta) if no image color is defined

Feature 9: Holiday Markers

- **Data source:** A static JSON array of US federal holidays for the current year (hardcoded — no API)
- **Visual:** Dates that are holidays show a small colored dot (6px) above the date number, in a distinct color (`#e63946` red)
- **Tooltip:** Hovering over a holiday date shows a tooltip with the holiday name, e.g., "Independence Day"
- **Holiday list (2025):** New Year's Day (Jan 1), MLK Day (Jan 20), Presidents' Day (Feb 17), Memorial Day (May 26), Independence Day (Jul 4), Labor Day (Sep 1), Columbus Day (Oct 13), Veterans Day (Nov 11), Thanksgiving (Nov 27), Christmas Day (Dec 25)

Feature 10: Page-Flip Month Transition Animation

- **Style:** A CSS-driven 3D flip or smooth slide animation when navigating months

- **Duration:** 300–400ms
- **Implementation:** CSS transform with `perspective(1000px)` and `rotateY` for flip effect OR horizontal slide with opacity — choose whichever performs better at 60fps
- **Reduced motion:** If `prefers-reduced-motion: reduce` is detected, instantly swap content with no animation

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📱 Screen Inventory

Screen 1: Desktop View (≥1024px)

- Single centered card, max-width 960px
- Top section: Hero image (full width of card, aspect-ratio 16/9, max-height 420px)
- Spiral binding decoration along top edge of image
- Month/year badge overlay on hero image (bottom-right)
- Below hero: two-column layout
  - Left column (60%): Month navigation bar + 7-column date grid
  - Right column (40%): Notes panel (monthly note + conditional range note)
- Dark mode toggle: top-right of card
- Total card height: approximately 780–850px depending on content

Screen 2: Tablet View (769px–1023px)

- Same layout as desktop but max-width shrinks to 90vw
- Hero image aspect-ratio adjusts to 3/2
- Date grid cells shrink proportionally
- Notes panel text area reduces height

Screen 3: Mobile View (≤768px)

- Full-width card with 16px horizontal margin
- Vertically stacked: Hero image → Navigation → Date grid → Notes panel

- Hero image: full width, aspect-ratio 16/10, max-height 240px
- Date grid: full width, minimum 40px cell sizes maintained
- Notes panel: full width, text areas stack vertically
- Day note popover: appears as a bottom sheet instead of a tooltip

Screen 4: Day Note Popover (Overlay)

- **Desktop:** 240px wide tooltip-style popover, appears above or below the clicked date cell
- **Mobile:** Bottom sheet that slides up from the bottom, 100% width, max-height 200px
- Contains: Date label, text input (140 chars max), save and delete buttons
- Backdrop: semi-transparent overlay on mobile only

🔄 Key User Flows

Flow 1: Select a Date Range

1. User views the calendar on the current month
2. User clicks on June 5 → cell fills with accent color (start date state), label appears in notes panel: "Start: Jun 5"
3. User clicks on June 12 → cell fills with accent color (end date state), all cells June 6–11 show the in-range band
4. Notes panel updates to show range note input: "Note for Jun 5 – Jun 12"
5. If user clicks June 3 (before start) → June 3 becomes start, June 5 becomes end, range re-renders between June 3–5
6. If user clicks June 20 while range (5–12) exists → range clears, June 20 becomes new start

Flow 2: Write a Monthly Note

1. User sees the notes panel with an empty text area labeled "Notes for June..."
2. User types "Plan team offsite"

3. Character counter updates: "17 / 500"
4. After 500ms of no typing, note auto-saves to localStorage
5. User navigates to July → notes area clears, shows "Notes for July..."
6. User navigates back to June → "Plan team offsite" reloads from localStorage

Flow 3: Write a Range Note

1. User selects June 5–12 as a range
2. Below the monthly note, a range note input appears: "Note for Jun 5 – Jun 12"
3. User types "Beach vacation with family"
4. Note auto-saves keyed to `"2025-06-05_2025-06-12"`
5. User clears the range → range note input hides
6. User re-selects June 5–12 → "Beach vacation with family" reloads

Flow 4: Add a Day Note

1. User double-clicks (or long-presses on mobile) June 15
2. Popover appears with text input: "Note for Jun 15"
3. User types "Dentist at 3pm"
4. User clicks "Save" → popover closes, a 4px dot appears below the "15" in the grid
5. User hovers over the dot → tooltip previews "Dentist at 3pm"
6. Double-clicking June 15 again re-opens the popover with saved note, user can edit or delete

Flow 5: Navigate Months

1. User clicks the right arrow (→) next to "June 2025"
2. Calendar grid slides out left and fades, "July 2025" grid slides in from right
3. Hero image crossfades to July's image
4. Accent color transitions to July's dominant color (e.g., deep green for a forest image)

5. Notes panel loads July's monthly note (or empty state)
6. Any previously selected range is still visible if it includes dates in July

Flow 6: Toggle Dark Mode

1. User clicks the sun/moon icon in the top-right
2. All colors transition over 300ms: background, card surface, text, borders
3. Hero image remains unchanged (only UI chrome changes)
4. Preference saves to localStorage as `wallcal_theme: "dark"`
5. On next visit, dark mode loads automatically

Flow 7: View a Holiday

1. User navigates to December 2025
2. December 25 shows a red dot (6px) above the "25"
3. User hovers over the date → tooltip appears: "Christmas Day"
4. Holiday dates are still selectable for ranges — the holiday indicator coexists with selection states

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📊 Success Metrics

Metric	Target	How to Verify
Lighthouse Performance Score	≥ 90	Run Lighthouse in Chrome DevTools
Lighthouse Accessibility Score	≥ 90	Run Lighthouse in Chrome DevTools
First Contentful Paint	< 1.5s	Lighthouse or WebPageTest
Animation frame rate	Consistent 60fps on transitions	Chrome DevTools Performance tab
Responsive breakpoints working	3 breakpoints render correctly	Manual test at 375px, 768px, 1280px
localStorage persistence	Data survives refresh	Refresh browser, verify notes reload
Range selection accuracy	All 5 selection behaviors work	Manual walkthrough of Flow 1

Metric	Target	How to Verify
Cross-browser	Works in Chrome, Firefox, Safari	Manual test in each browser
Mobile touch	Range select + long-press notes work on touch	Test on real device or DevTools emulator
Code quality	No TypeScript errors, no ESLint warnings	`npm run build` passes clean

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🚫 Out of Scope

- **No backend / database / API calls** — strictly frontend, localStorage only
- **No recurring events or event scheduling** — this is a date range selector, not a full calendar app
- **No multi-calendar support** — single calendar instance
- **No collaboration / sharing / multi-user** — single-user, local-only
- **No Google Calendar / iCal / external integration**
- **No drag-to-select ranges** — click-based selection only (drag adds complexity without proportional UX value)
- **No year view or week view** — month view only
- **No print stylesheet** — screen-only
- **No i18n / localization** — English only, US holidays only
- **No image upload** — hero images are static/predefined
- **No export/import of data** — deferred to post-MVP if time allows

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🎯 Development Phases

Phase 1: Foundation (Days 1–2)

☐ Project setup: Next.js + TypeScript + Tailwind CSS

☐ Calendar grid component: render correct days for any month with proper weekday alignment

☐ Month navigation: prev/next with month/year display

- [ ] Basic styling: card layout, grid spacing, column headers
- [ ] Responsive layout: desktop two-column, mobile stacked

Phase 2: Core Interactions (Days 3–4)

- [ ] Date range selection: implement the 5-behavior selection model
- [ ] Visual states: start, end, in-range, hover, today — all 5 states styled
- [ ] Monthly notes text area: functional with character counter
- [ ] Range notes: appear/disappear based on active selection
- [ ] localStorage: save and load all data types

Phase 3: Visual Polish (Days 5–6)

- [ ] Hero image integration: 12 month-specific images, proper sizing
- [ ] Spiral binding CSS decoration
- [ ] Month/year badge overlay on hero
- [ ] Month transition animation (slide + fade or flip)
- [ ] Accent color tinting per month
- [ ] Dark mode toggle + color transition

Phase 4: Extras & QA (Days 7–8)

- [ ] Day notes: double-click/long-press popover
- [ ] Holiday markers: static data + dot indicators + tooltips
- [ ] Accessibility audit: keyboard navigation, aria labels, focus states
- [ ] Cross-browser testing: Chrome, Firefox, Safari
- [ ] Mobile testing: touch interactions, bottom sheet for day notes
- [ ] Reduced motion support
- [ ] Lighthouse audit and performance optimization

Phase 5: Delivery (Days 9–10)

- [ ] Deploy to Vercel
- [ ] Write README.md with architecture decisions, setup instructions, screenshots

- [ ] Record Loom video demonstrating all features
- [ ] Final code cleanup: remove unused imports, add JSDoc comments to key functions
- [ ] Push to public GitHub repository

🔐 Privacy & Safety

- **No user data leaves the browser** — all data stored in localStorage, never transmitted
- **No cookies** — zero tracking
- **No analytics** — no third-party scripts
- **No authentication** — no user accounts
- **Content Security:** Notes are plain text only, rendered with `textContent` (not `innerHTML`) to prevent XSS
- **localStorage limits:** Handle `QuotaExceededError` gracefully — show a toast notification, don't crash
- **Image loading:** All hero images loaded from static assets or Unsplash with proper `alt` text for accessibility

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✅ Definition of Done

The component is "done" when ALL of the following are true:

1. ✅ Calendar renders the correct grid for any month (verified for Feb 2024 leap year, months starting on different weekdays)
2. ✅ Date range selection works per all 5 defined behaviors with correct visual states
3. ✅ Monthly notes save and reload per month via localStorage
4. ✅ Range notes save and reload for specific date ranges via localStorage
5. ✅ Day notes work via double-click/long-press with dot indicators in the grid
6. ✅ Month navigation works with transition animation

7. ✅ Hero image changes per month with accent color tinting
8. ✅ Dark mode toggle works and persists preference
9. ✅ Holiday markers display correctly with tooltips
10. ✅ Responsive layout works at 375px (mobile), 768px (tablet), 1280px (desktop)
11. ✅ `prefers-reduced-motion` disables animations
12. ✅ Keyboard navigation: Tab through interactive elements, arrow keys for month nav
13. ✅ `npm run build` completes with zero errors and zero warnings
14. ✅ Lighthouse Performance ≥ 90, Accessibility ≥ 90
15. ✅ Deployed to Vercel with a working public URL
16. ✅ Video demo recorded showing all features
17. ✅ README.md documents architecture, setup, and design decisions

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🎨 Design System

Typography

Element	Font	Size	Weight	Line-height
Month/Year overlay	System sans-serif (Inter or Geist)	20px	700	1.2
Day column headers	System sans-serif	11px	600	1
Date numbers	System sans-serif	14px	400 (500 for today)	1
Notes label	System sans-serif	11px	600	1
Notes body text	System sans-serif	14px	400	1.6
Character counter	System sans-serif	11px	400	1

Element	Font	Size	Weight	Line-height
Navigation month/year	System sans-serif	18px	600	1.2

Spacing Scale

Token	Value
`space-xs`	4px
`space-sm`	8px
`space-md`	16px
`space-lg`	24px
`space-xl`	32px
`space-2xl`	48px

Color Tokens (Light Theme)

Token	Value	Usage
`--bg-primary`	`#FFFFFF`	Page background
`--surface`	`#FFFFFF`	Card background
`--surface-secondary`	`#F8F8F8`	Notes panel background
`--text-primary`	`#1A1A1A`	Main text
`--text-secondary`	`#6B6B6B`	Labels, counters
`--text-muted`	`#B0B0B0`	Overflow days, placeholders
`--border`	`#E5E5E5`	Dividers, panel borders
`--accent`	Dynamic per month, default `#E07A5F`	Selection, weekends, today
`--accent-10`	Accent at 10% opacity	Hover state
`--accent-15`	Accent at 15% opacity	In-range band
`--holiday`	`#E63946`	Holiday dot indicator
`--shadow`	`0 8px 32px rgba(0,0,0,0.12)`	Card elevation

Color Tokens (Dark Theme)

Token	Value	Usage
`--bg-primary`	`#0F0F1A`	Page background
`--surface`	`#1A1A2E`	Card background
`--surface-secondary`	`#252540`	Notes panel background
`--text-primary`	`#E8E8E8`	Main text
`--text-secondary`	`#9A9AB0`	Labels, counters
`--text-muted`	`#4A4A60`	Overflow days, placeholders
`--border`	`#2A2A40`	Dividers, panel borders
`--shadow`	`0 8px 32px rgba(0,0,0,0.4)`	Card elevation

Monthly Accent Colors & Hero Images

Month	Accent Color	Hero Image Theme
January	`#457B9D` (steel blue)	Snow mountain landscape
February	`#E07A5F` (terracotta)	Warm sunset
March	`#81B29A` (sage green)	Spring blossoms
April	`#F2CC8F` (golden yellow)	Rainy field
May	`#2A9D8F` (teal)	Lush forest
June	`#E9C46A` (sunflower)	Beach/ocean
July	`#264653` (deep teal)	Night sky / fireworks
August	`#F4A261` (warm orange)	Golden hour hills
September	`#C17817` (amber)	Autumn foliage
October	`#D64933` (burnt red)	Fall harvest
November	`#6D597A` (muted purple)	Foggy morning
December	`#355070` (navy)	Snowy village

Border Radius Scale

Token	Value
`--radius-sm`	4px
`--radius-md`	8px
`--radius-lg`	12px
`--radius-full`	9999px

Animation Tokens

Token	Value	Usage
`--transition-fast`	150ms ease	Hover states
`--transition-base`	200ms ease-out	Selection state changes
`--transition-slow`	300ms ease-in-out	Month transitions, theme toggle
`--transition-accent`	400ms ease	Accent color crossfade

Component-Specific Specs

Component	Spec
Calendar card	max-width: 960px, border-radius: 12px, shadow: `--shadow`
Hero image	aspect-ratio: 16/9 (desktop), 16/10 (mobile), object-fit: cover, border-radius: 12px 12px 0 0
Spiral binding	height: 24px, repeating CSS radial-gradient or pseudo-elements, positioned at top of hero
Grid cell (desktop)	48×48px min, centered text
Grid cell (mobile)	40×40px min, centered text
Nav arrow buttons	36×36px, border-radius: 50%, ghost style
Notes text area	border: none, border-bottom: 1px solid `--border`, resize: vertical, min-height: 80px
Day note dot	4px × 4px, border-radius: 50%, centered below date number
Holiday dot	6px × 6px, border-radius: 50%, `--holiday` color, centered above date number
Tooltip	padding: 6px 12px, border-radius: 6px, font-size: 12px, background: `--text-primary`, color: `--surface`
Dark mode toggle	32×32px button, top-right corner, 8px from edges